xmitter Rev D Pre-Layout Design Review

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Scope and ground truth

Schematic-only review of the xmitter Rev D branch before any of the boards go to layout. Ground truth is the five netlists (`analog.net`, `ss_pa.net`, `Output LPF.net`, `frontpanel.net`, `protoshield.net`); the handoff narrative is used for intent only. Several of the narrative's claims describe circuitry that is not in the netlists; those are flagged where they matter.

The seven items that most need attention: the Metro's pins are absent from the protoshield netlist, the LM7171 output is DC-shorted through T201, the DGFET cannot cut off at key-up, the ss_pa MCP1700 is fed 13.8 V, there is no fuse or reverse-polarity protection anywhere, the I²C bus is pulled to 5 V onto 3.3 V parts, and the IRF510 bias-kill is ~10 ms rather than the documented 18 µs.

Must-fix before layout

Anything here will produce a dead board, a damaged part, or a transmitter that does not key off.

M1 — protoshield: the Metro's pins are not in the netlist (downgraded to documentation if the shield is hand-wired). `Metro_ESP32S3` appears in `libparts` but there is no `U2` component and no net touches a Metro pin, so ENC_INT_FINAL, CS_DAC, MOSI, SCK, FAULT_N and LDAC_4728 are single-node or resistor-only nets and `+3.3V` has no source. The shield plugs into the Metro, but that only connects the header pins; the schematic still has to say which header pin each conditioned signal lands on. If the shield is the Adafruit #2077 perfboard, hand-wired, this is a documentation gap: place the Metro symbol and wire it so the GPIO assignments (D2/FREQ_A, D4/DIT, D10/CS_DAC, etc.) and the 3V3 feed for the pull-ups are recorded. If it will be a fabricated PCB, the Metro header footprint must be in the netlist or the board has no copper to those pins.

M2 — analog/ss_pa: LM7171 output is DC-shorted through T201. U1 is a DC-coupled follower on the DGFET drain (~+6 V, rising toward +12 V at key-up). Its output runs `R2 50R → J1 → ss_pa J3 → T201 primary → GND` with no blocking cap: ~100 mA current-limit, ~1

W in a DIP-8, and a DC-magnetised BN61 core. Fix: AC-couple the drain into U1 (series C into R1, bias pin 3 with a resistor) and add a series C at the output. As drawn the key-up drain voltage also exceeds the LM7171 input common-mode limit (V+ − 1.5 V).

M3 — DGFET cannot cut off at key-up (backwave). Q2 source is tied to GND, G1 sits at 0 V (R21 1 M), and the Q11 mute can only pull G2 to ~0 V. A 40673 at V_G2S = 0 gives roughly 25–35 dB attenuation, not a keyed-off carrier. Add a bypassed source resistor (≈220–270 Ω + 100 nF) so key-up puts both gates about −1.5 V relative to source. Note the netlist has MCP4922 **VoutB → R13 → Gate 2** and **VoutA → DRIVER_BIAS**, the opposite of the handoff; firmware must match.

M4 — ss_pa U13 MCP1700 fed from +13.8 V. MCP1700 absolute-max V_IN is 6 V; it dies at power-on. Use MCP1703-3302 (16 V max), LM2936 or 78L33. Analog U13 and frontpanel U4 are on +5 V and are fine.

M5 — no fuse or reverse-polarity protection on the shack input. The handoff cites a D1 Schottky and a D5 blocking diode; neither exists. Analog D5 (1N4001) is the K5 coil flyback; J5 lands on the raw bus. ss_pa D2 (1N5822) parallels L5 as a freewheel, not a series blocker. Add a fuse at J5 and a series Schottky or P-FET. This is the one safety item.

M6 — I²C bus is pulled to 5 V onto the ESP32-S3. The PCF8575 and MCP4725 breakouts are rated for 5 V VIN; that is not the problem. Adafruit's PCF8575 guide states SDA/SCL use the same logic level as VIN and carry 10 k pull-ups to it, and the MCP4725 breakout has the same arrangement to its VDD. With analog U17, frontpanel U1 and U2 all on +5 V, the shared bus idles at 5 V, and the Metro ESP32-S3 I²C pins are 3.6 V absolute max. Two ways out: (a) power the three breakouts from +3.3 V (the frontpanel LDO currently powers only one pull-up, so that rail is already there; on analog, run U18 CD14538B from 3.3 V too or the heartbeat trigger fails CD4000 V_IH of 0.7·VDD, and move R51 to 3.3 V), or (b) keep them at 5 V and put a BSS138-style bidirectional level shifter between the Metro and the bus. Option (a) is fewer parts; the 3.3 V PCF8575 still meets the HD44780's 2.2 V V_IH and the 3.3 V MCP4725 covers the LCD V0 range.

M7 — MCP4922 at 5 V driven by 3.3 V SPI. MCP4922 V_IH = 0.7·VDD = 3.5 V. Power U4 from +3.3 V (VREF ⩽ 3.3 V is still ample for G2 and BS170 bias) or level-shift. 3.3 V is the cleaner fix and matches M6.

M8 — IRF510 bias range, and the bias-kill is 10 ms, not 18 µs.

- U3 (MCP4728) VIN and U5 (Si5351) VIN are marked no-connect. If both are powered over STEMMA QT from the Metro, the DAC is on 3.3 V, so PA_Bias ⩽ 3.3 V minus HC4066 drop. IRF510 V_GS(th) is spec'd 2–4 V; the ~100 mA class-AB point is typically

3.3–3.8 V. Either power U3 at 5 V and use a **74HCT4066** (V_IH 2 V, so 3.3 V FAULT_N still controls it at VCC = 5 V; needs a 78L05 on ss_pa beside the 3.3 V LDO), or accept 3.3 V and hand-select devices.

- The 4066 isolates the DAC, but the gate node still has C9/C10 (0.1 µF) discharging only through R12/R19 100 k: τ = 10 ms. Reduce R12/R19 to 1–2.2 k or add a shunt-kill FET.
- Killing bias does not stop RF: the BS170s are still driven and the IRF510s conduct on gate RF peaks in class C until firmware also mutes the envelope.

M9 — ss_pa fault comparator sees raw 14 MHz. Source_Sense_A/B run straight from the 0.11 Ω source node to the LM393 inputs with no filtering (C11 is on the output). The comparator trips on RF peaks, and the V_REF ≈ 0.30 V threshold (2.7 A per device) is above anything useful on average. Add 1 k + 10 nF at each −in, send the filtered node over the umbilical rather than raw RF, and re-pick R10 (~5.1 k → ~0.165 V → ~1.5 A average per device).

M10 — fault latch has zero margin. After a trip FAULT_N → 0 V pulls V_REF to 0 V; the −in also sits at 0 V once the PA stops, so latch state is decided by LM393 offset (±2–5 mV). Give the −in a small positive floor (220 k from +3.3 V → ~15 mV with the 1 k filter).

M11 — frontpanel umbilical has no ground conductor. Analog J4 and frontpanel J1 have GND only on `SH`. All backlight, encoder and logic return (~150 mA) rides the cable shield; an unshielded patch cable has no return at all. Put GND on a pin (pin 5 frees up under M13), ideally paired with +5 V.

M12 — frontpanel backlight FETs cannot turn on. R1/R2/R3 sit in the **source** leg of Q1–Q3. At ~14 mA through 120–200 Ω the source is at 2–3 V; a 3.3 V gate gives V_GS ≈ 0.5 V (off), and at 5 V brightness depends on V_GS(th) spread. Move the resistors to the drain side. R2/R3 are not RS-422 terminators as the handoff says; the real 120 Ω terminators are R100/R101 on analog, correctly at the receiver. RED/GREEN/BLUE are driven by the frontpanel PCF8575 P8–P10, not the Metro.

M13 — ENC_INT has no driver; frontpanel J4 is four pins on one net. The MBL-600 has no interrupt output. `ENC_INT_RAW` is sourced only by J4 (all four pins shorted, no GND) and a 10 k pull-up, while PCF8575 IRQ is no-connect. Either delete the net or wire PCF8575 IRQ (open-drain) to it; R4 10 k to 3.3 V is then exactly right and gives the Metro a button-press interrupt.

M14 — encoder quadrature path is too slow and the level translation is wrong. AM26LS32 (5 V TTL) → R26/R27 2.2 k → R28/R29 10 k pull-down → protoshield R13/R14 1 k → 100 nF + 1N4728A → Metro. Thevenin ~3 k into 100 nF ≈ 300 µs per edge; a brisk spin of a

100 PPR encoder is 200–500 µs per edge. The 1N4728A at ~0.3 mA is a soft, leaky clamp. Make R28/R29 ≈ 3.3 k (3.0 V high), drop C8/C9 to ≤1 nF or remove, drop D3/D4.

M15 — LPF relay contact function vs. symbol text. The `Relay_G6K` description says 2/3/4 = pole 1 NC/COM/NO and 5/6/7 = pole 2 NC/COM/NO. The wiring uses pin 4 and pin 5 for the RF busses and pins 2 and 7 for the swamps, i.e. it assumes 4 ≡ 5 and 2 ≡ 7 in function. The G6K-2P is physically symmetric, so the wiring is probably right and the text wrong — but if 4/5 are NC, every bank is in-circuit at rest and energising swamps it. Confirm against the datasheet bottom-view drawing before fab.

Should-consider

Non-blocking, but each removes a real risk or a part.

S1 — eliminate the −12 V rail. U1′s input is already near mid-rail by accident; do it on purpose (AC-couple in, bias pin 3 at ~+6 V, AC-couple out) and run the LM7171 single-supply from +12 V. That removes the LT1054, its 25 kHz switching noise, and the headroom problem: with R16 20 k / R17 240 k the regulated target is ~16 V, unreachable from 13.8 V, so it runs open-loop at ≈ −(V_IN − 1.1 V). The LT1054 is also rated 100 mA, about what the op-amp′s negative rail draws when keyed.

S2 — driver headroom and dissipation. Through R2 50 Ω into 50 Ω the LM7171 delivers about ±5 V at the load (≈0.25 W) at 100 mA peaks, with ±12 V rails and 100 °C/W in a DIP-8; it will run hot. Estimated chain: DGFET ≈ 3–4 V pk at the drain → ~40 mW into the PA at unity gain. Check this against the QRP Labs PA′s actual drive requirement before relying on the gain-4 R_g hedge.

S3 — the on-analog Chebyshev LPF and the LM7171 push-pull driver (U4/U5) in the handoff do not exist. The netlist has one LM7171 (U1); U4 is the MCP4922 and U5 the Si5351. The Si5351 square wave reaches the PA with its 3rd harmonic (42 MHz) only ~10 dB down (R1/C1 helps ~3 dB). The output LPF handles it at the antenna, but PA gain and heat are spent on harmonics. A 2-pole LC at the DGFET drain is cheap.

S4 — PA return current through the analog board. ss_pa +13.8 V comes via analog K5 → J33 and returns via J33 pin 1 → analog GND → J5: 2–3 A through the analog ground pour next to the DGFET and op-amp. Return the PA supply directly to the shack-supply negative, or at least a dedicated heavy trace star-pointed at J5. Size K5 and the bus trace for it.

S5 — ss_pa C6 (4700 µF) inrush. R7 750 Ω on K5′s NC contact is a pre-charge, but only if J34 is wired in parallel with J33 at the ss_pa; document that. τ = 3.5 s, so firmware must not close K5 for ~10 s after power-up, or reduce R7 (100 Ω / 2 W → 0.5 s).

S6 — U6 LM2940-5 thermal. The +5 V load is roughly AM26LS32 (~70 mA), LCD + backlight (~50 mA), MBL-600 (~50–100 mA), two PCF8575s, MCP4922/4725, plus J13. At ~0.4 A, (13.8 − 5) × 0.4 ≈ 3.5 W in a bare TO-220: needs a real heatsink or a buck module. Same for +12 V: if J21 (polyfuse → +12 V) feeds the Metro barrel jack, the handoff′s 75–100 mA estimate is low by a few hundred mA.

S7 — PCF8575 power-on state is all-HIGH. On analog that asserts RLY_15/17/20/40 together (four banks in parallel, ~47 mA of coils), P15 TX/RX on J6, and P14 PA power (gated by the heartbeat, so safe). Make HIGH the safe polarity on every port, or run the LPF relays active-low. The LPF 2N7000 gates come straight off the RJ45 with no pull-down or series R: add 100 k + 1 k. J6 (T/R relay output, Q4 drain) has no flyback diode.

S8 — analog↔protoshield terminal blocks do not mirror. J20 (GND/SCK/MOSI/CS_DAC) mates to shield J7 (CS_DAC/MOSI/SCK/NC) in reverse order; J12 splits across shield J1/J2; J22 across J3/J5; J14/J3 versus J8. Not electrically wrong, but a guaranteed miswire. Regularise to 1:1 pin order and produce a harness table. `PCF_P5_SPARE` (J14.4) is a dangling single-node net not connected to U17.

S9 — paddle inputs. Only 10 k pull-ups on PADDLE_A/B: no series R, cap or clamp (R1/R2 1 k are on Source_Sense). For a transmit environment: 4.7 k pull-up, 1 k series, 10 nF at the terminal. Same for FAULT_N (shield J8.3 goes straight to the GPIO) and TR_SENSE.

S10 — Source_Sense has no clamp. The handoff puts 1N4728A on Source_Sense_A/B; the netlist has D1/D2 on I_CATHODE, D3/D4 on MBL600, D5 on GRID_BLOCK_CRASH. With the ss_pa-side filter from M9 the shield RC is enough; update the doc.

S11 — SDA/SCL on one twisted pair (J4 pins 1/2). Classic I²C-over-cable crosstalk; pair each with GND if possible and keep the bus at 100 kHz.

S12 — IRF510 tabs are the drains, at opposite RF phase: insulated mounting on the shared sink is mandatory. Thermal budget (~4 W/device, ~40 °C rise on the QRP Labs sink at CW duty) is fine.

Nice-to-have

- ss_pa J5 (+3.3 V / V_REF) is already a latch-reset header; add a momentary button and the fault clears without a power cycle.

- Frontpanel J2 (P13–P15 buttons) has no GND pin.
- Protoshield R7 is 330 Ω (handoff says 1 k); R8 is DNP.
- LPF inductor `Description` fields say T68-6 while footprints and the handoff say T50-6.
- LPF caps: shunt elements see ~0.8 A RF and 32 V pk (64 V into an open); specify 100 V C0G.
- The 40673 is long out of production; keep spares or note BF981/3N201 as TO-72 substitutes.
- CF7/CF8/CF12/CF16 "2 in parallel" notes contradict their single values; clean up before BOM export.
- MCP4922 ~LDAC is on a PCF8575 pin; for the 25 µs envelope loop hold it low permanently (or move it to a Metro GPIO) so SPI writes update immediately.

Answers to the handoff's review questions

Question	Answer
LPF values and isolation	All four banks match 7-pole 0.1 dB Chebyshev g-values (1.1812 / 1.4228 / 2.0967 / 1.5734) within E12 rounding; C-L-C-L-C-L-C topology is correct in every bank. 20 m 2nd harmonic at 28 MHz is 1.64·fc → ~43 dB from the filter plus push-pull cancellation; 3rd harmonic >70 dB. −43 dBc is met with margin. T50-6 is fine at 10 W.
Shared vs per-bank swamp	Per-bank is adequate. During break-before-make the PA sees an open; only matters if keyed during a band change. Firmware interlock, no hardware.
Fault reset policy	Keep the latch; add the button on J5. Do not give firmware a clear path.
Heartbeat timing	R30/C41 give ~0.47 s; retrigger at ≥3 Hz. Triggering via the PCF8575 means an I²C hang also drops the PA, which is a feature. The Q5/Q7 stack correctly ANDs "firmware wants PA" with "firmware is alive".
+12 V budget	Fine for the LM2940 if the Metro is not on it (see S6).

Question	Answer
G6K-2P-Y-T1 vs G6K-2P	Electrically identical for this duty (12 V coil, same pinout); suffixes are variant/packaging. Any 12 V G6K-2P works. Do not hot-switch the contacts at 10 W.
MCP4728 address reprogram	The Si5351 shares 0x60 during the one-time write and sees the command bytes as register writes. Harmless if done before Si5351 init, then reset the Si5351, or unplug its STEMMA cable for that step.
RS-422 termination placement	R100/R101 on analog are correct (receiver end). Frontpanel R2/R3 are backlight resistors (M12).
Frontpanel 3.3 V	Local-only and currently powers one pull-up; it should power U1/U2 (M6). No parallel-rail conflict.
Zener clamps	1N4728A at sub-mA is a poor logic clamp (soft knee, ~100 µA leakage near Vz). Dividers for the 5 V TTL lines; series R plus ESP32 internal diodes for the rest.
Firmware factoring	Core-1 pinned GPTimer ISR at 25 µs writing the MCP4922 over SPI is the right shape; keep I²C off core 1. Heartbeat and band select over the PCF8575 are fine.
Metro power	Nothing in the netlists defines it. J21 (polyfuse → +12 V) looks like a barrel-jack feed; confirm, and add the Metro's regulator to the +12 V budget.
IRF510 thermal	~4 W per device, ~40 °C rise on the QRP Labs sink at CW duty: adequate. Insulated mounting is mandatory (S12).

Bottom line

Fix M1–M15, then re-export the netlists and re-review the three touched areas (op-amp coupling, bias/fault chain, logic supply voltages) before layout. The must-fix list is dominated by rail-voltage mismatches between 5 V and 3.3 V parts and by narrative claims that never made it into the schematics; both are cheap to fix now and expensive after fab.